Probability & Statistics 101

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Preface

This is an Introduction to Probability and Statistics book written with Quarto for the GA!

To learn more about Quarto books visit https://quarto.org/docs/books.

Note: This is a work-in-progress and expect revisions and updates intermittently!

1 Introduction

Learning objectives

By the end of this chapter, you should be able to:

- explain what statistics is and why it is useful,
- distinguish clearly between data and information,
- describe the difference between descriptive and inferential statistics,
- explain the concepts of population, sample, parameter, and statistic,
- understand why inference requires probability,
- recognize the classical and Bayesian approaches to statistics.

How to use these notes

This book is written to be read **narratively**, not just consulted for formulas. Each chapter introduces ideas gradually, motivating statistical concepts through explanation, examples, and visual intuition before formal definitions appear.
You are encouraged to: - read the text carefully before focusing on formulas, - study graphs and numerical examples together, - pause at highlighted boxes to reflect on key ideas and common pitfalls, - attempt the end-of-chapter exercises to check and deepen your understanding.
Later chapters build on earlier ones, so a solid grasp of the foundational concepts in descriptive statistics and probability will make the material on correlation and statistical inference much easier to follow.
Statistics is best learned by **thinking with data**, not by memorizing techniques. This book is designed to support that process.

1.1 What is statistics?

Today's world is full of data. Data have become indispensable for making informed decisions, finding solutions to problems, understanding complex situations, and developing strategies that ultimately improve the lives of individuals and communities.[1]

[1] Think about the last decision you made using numbers—was it about money, health, grades, or something else?

Although the terms *data* and *information* are often used interchangeably, they have distinct meanings. **Data** can be regarded as facts—especially numerical facts—that are collected for reference or analysis. **Information**, on the other hand, is knowledge communicated about some particular fact or phenomenon of interest.

> **Big picture**
>
> Data by themselves are often messy and overwhelming. Statistics provides the tools that turn raw data into meaningful **information**.

Put simply, **statistics** is a **tool** for extracting **information from data**. What the consumer of statistics ultimately seeks is information—something that helps them see through and make sense of the maze and dazzle of raw data.

Statistics helps us create new understanding from data. Broadly speaking, there are two main branches of statistics:

- **Descriptive statistics**, which focus on organizing, summarizing, and presenting data in a clear and informative way.
- **Inferential statistics**, which use data from a sample to make conclusions or draw inferences about some unknown aspect of a population.

> **Key idea**
>
> **Descriptive statistics** describe *what the data look like.*
> **Inferential statistics** help us learn about *what we cannot directly observe.*

1.2 Statistical inference

Descriptive statistics rely on graphical summaries and numerical techniques. **Statistical inference**, in contrast, involves making estimates, predictions, and decisions about a population based on information obtained from a sample.

In practice, what a statistician is often interested in is a **population**—a collection of all possible individuals, objects, or measurements of interest. Of particular importance is some numerical characteristic of the population, known as a **parameter**.

Because it is usually impossible (or impractical) to observe the entire population, a **sample** is drawn. From this sample, a **statistic**—a numerical quantity describing some feature of the sample—is calculated. Statistical inference then uses this statistic to say something about the corresponding population parameter.

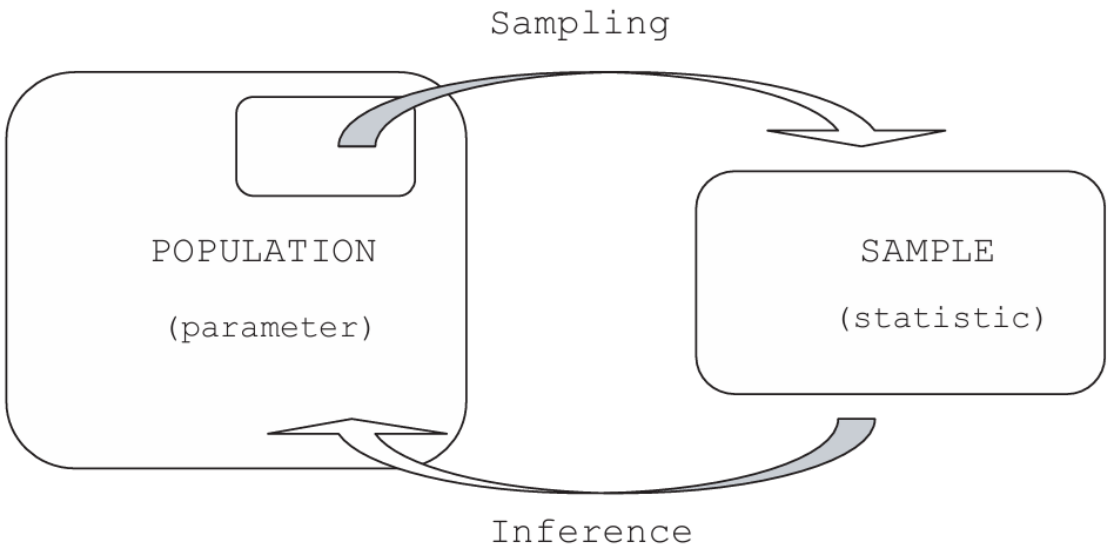


Figure 1.1: Statistical Inference

Before we treat inferential statistics later in this book in detail, we begin with a discussion of basic statistical concepts that are best introduced through descriptive statistics, together with an introduction to elementary probability, which is essential for understanding statistical inference.

1.3 Pioneers of statistics

There are several approaches to doing statistics. In this book, we primarily adopt the **classical approach**, developed largely by **Sir Ronald Aylmer Fisher (1890–1962)**—a British statistician, geneticist, and professor. For his contributions, Fisher has been described as *"a genius who almost single-handedly created the foundations for modern statistical science."* (Efron 1998)

Another important framework is the **Bayesian approach**, associated with **Thomas Bayes (1702–1761)**, an English statistician, philosopher, and Presbyterian minister. Bayes is best known for formulating what is now called **Bayes' theorem**, which will be introduced in a later chapter.[2]

[2]Interestingly, Bayes never published what would become his most famous accomplishment; his notes were edited and published after his death by Richard Price.

1.4 Key terms

1.4.1 Data

Data are characteristics or pieces of information—usually numerical—collected through observation. More formally, **data** are sets of values of qualitative or quantitative variables measured on one or more individuals or objects. A **datum** (the singular of data) refers to a single value of a single variable.

1.4.2 Statistics

Statistics is the discipline concerned with the **collection, organization, analysis, interpretation, and presentation of data**.

1.4.3 Descriptive statistics

Descriptive statistics refers to the process of **summarizing and describing data**, typically using tables, graphs, and numerical measures.

1.4.4 Inferential statistics

Inferential statistics (or statistical inference) is the process of **using data from a sample to draw conclusions about an underlying population or probability distribution**.

! Chapter summary

Statistics provides tools for turning raw data into meaningful information.
Descriptive statistics summarize and present data, while inferential statistics use data from samples to draw conclusions about populations. Central to statistical reasoning are the ideas of populations, samples, parameters, and statistics, as well as the role of probability in making inference possible.
This chapter sets the conceptual foundation for the rest of the book and motivates the need for the descriptive and inferential tools developed in later chapters.

2 Graphical Summaries

Learning objectives

By the end of this chapter, you should be able to:

- explain why graphical summaries are useful for understanding data,
- distinguish between cardinal, ordinal, and nominal data,
- identify appropriate graphical representations for different types of data,
- construct and interpret histograms correctly, including those with unequal class intervals,
- explain how histogram areas represent percentages or probabilities,
- recognize common histogram shapes such as symmetric, skewed, unimodal, and bimodal distributions.

A defining characteristic of today's information age is the abundance of data. Data in their raw form can be difficult to comprehend. One useful way to summarize a mass of data is through **graphical or pictorial representations**.

Many such graphs—pie charts, bar graphs, and so on—are easy to understand and commonly used. In this chapter, we focus on the **most important graphical representation in statistics: the histogram**.[1]{.sidenote}

Big picture

Graphs help us *see* patterns in data that are often invisible in raw numbers.

2.1 Types of data

Data are not all the same. A helpful way to think about them is in **three levels**:

- **Cardinal data**
- **Ordinal data**
- **Nominal data**

[1] Why might graphs be more informative than tables when data sets are large?

Once you know which kind of data you have, it becomes much easier to choose sensible statistics and graphs.

2.1.1 Cardinal data

Cardinal data (sometimes further divided into *interval* and *ratio* data) are represented by real numbers—such as heights, weights, and prices. These are **quantitative (numerical)** data, so arithmetic operations are meaningful.

For example, with heights or prices, it makes sense to say someone is *twice as tall* or that one item is *three times as expensive*.

2.1.2 Ordinal data

Ordinal data are values for which **only the order (ranking)** is meaningful. A common example is a course evaluation scale such as:

- Poor = 1
- Fair = 2
- Good = 3
- Very Good = 4
- Excellent = 5

Here, arithmetic operations usually do **not** make sense. It would sound strange, for instance, to say:

`2 × Fair = Very Good`.

Still, ordinal data are useful because the ranking is clear: *Excellent* (5) is better than *Very Good* (4), which is better than *Good* (3), and so on. Importantly, that ordering remains the same even if we changed the numbers used to label the categories.

2.1.3 Nominal data

Nominal data are usually **qualitative or categorical**. The numbers attached to categories serve only as labels. For example, marital status might be coded as:

- Single = 1

- Married = 2
- Divorced = 3
- Widowed = 4

Because these numbers are arbitrary, arithmetic operations are meaningless. It is absurd, for example, to say:

`married ÷ 2 = single`.

With categorical data, the main meaningful calculation is **counting**—how often each category occurs. We often summarize nominal data using a **frequency distribution**, or a **relative frequency distribution** showing proportions.

> ❗ Key idea
>
> The *type* of data determines which graphs and summaries make sense.

2.1.4 Why this matters

Keeping the data type in mind helps determine which statistical tools—graphical or otherwise—are appropriate. Among graphical techniques, we will emphasize the **histogram**, which plays a central role in statistics and probability.

2.2 The histogram

To understand how a histogram works, let us look at a concrete example.
Table 4.1 shows the **percentage of families** in each **income class interval**, measured in thousands of baht per year, for Bangkok residents in a recent year.

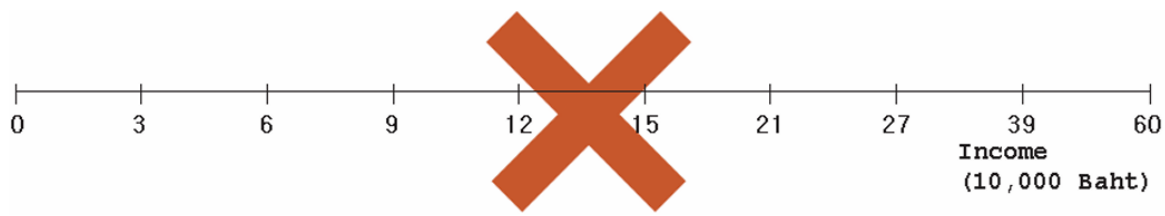
Table 2.1: Annual income for a sample of Bangkok residents

Income (Baht '000)	Percentage
0–30	1
30–60	3
60–90	6
90–120	10
120–150	11
150–210	24
210–270	20

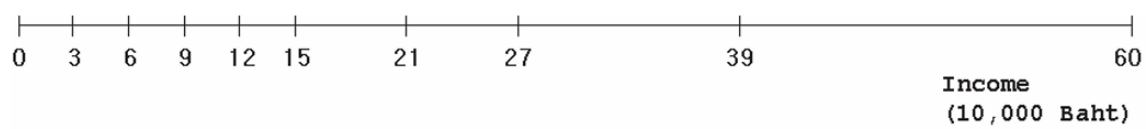
Income (Baht '000)	Percentage
270–390	18
390–600	5
Over 600	1

To construct a histogram from Table 4.1, the **first step** is to draw the **horizontal axis**, representing income.

A common mistake at this stage is to give **each class interval the same width**, regardless of its actual size. This leads to a misleading picture.

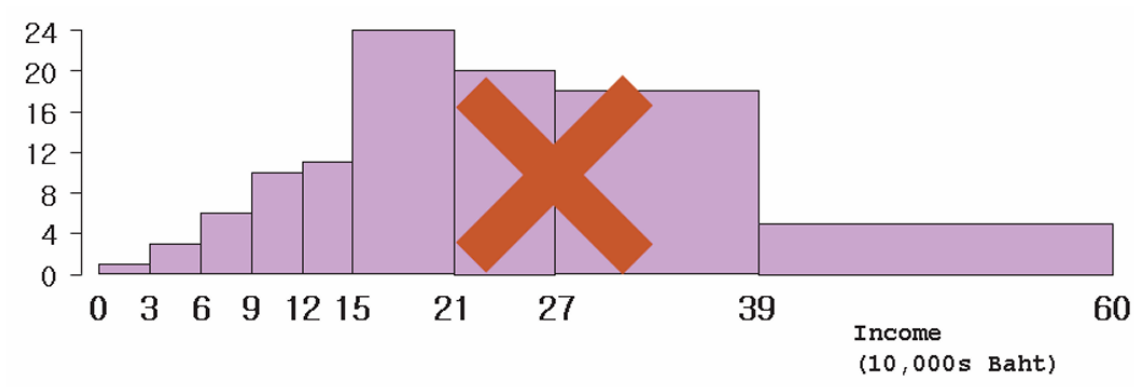


Instead, the horizontal axis should reflect the *actual width* of each income class, as shown below.[2]



The next step is to draw the blocks. Students often make the mistake of setting the height of each block equal to the percentage in the table. The figure below shows what happens if this is done.

[2]Class intervals need not be equal; wider intervals may be used where data are sparse.



This gives a misleading impression—for example, it appears that there are more families earning over 390,000 baht than under 120,000 baht, which is not correct.

> ⚠ Common pitfall
>
> In a histogram, **heights alone do not represent frequencies** when class widths differ.

Because the class intervals are unequal, the height of each block must be calculated as:

$$\text{percentage} \div \text{class width (relative length)}$$

For example, the interval 150–210 spans two 30,000-baht units. Dividing 24% by 2 gives the correct height. The vertical axis is therefore a **density scale** (percent per 30,000). The correct histogram is:

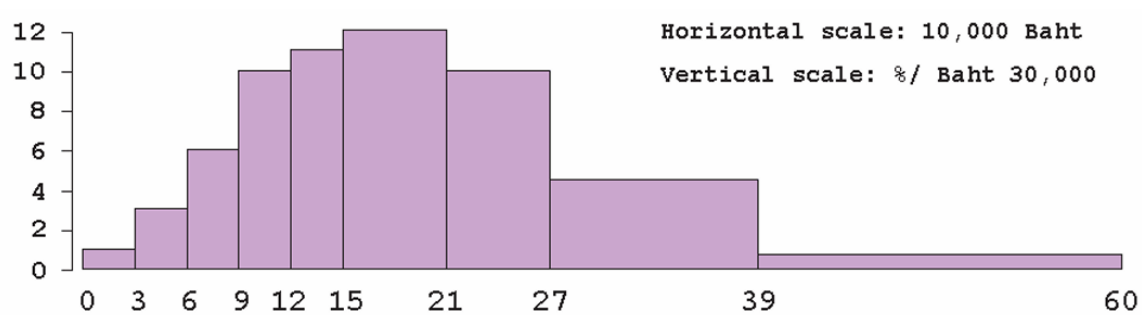


Figure 2.1: Correct Histogram

In a histogram: - **areas** represent percentages (or probabilities) - **heights** represent crowding per horizontal unit

An important property to remember is that the **total area under the histogram must equal 100 percent**.

> **!** Key idea
>
> Histograms represent data through *areas*, not just heights.

2.3 Shapes of histograms

Histograms come in many shapes. A histogram is **symmetric** if the two sides mirror each other around a central vertical line. The most famous symmetric histogram is the **Gaussian (normal) distribution**.

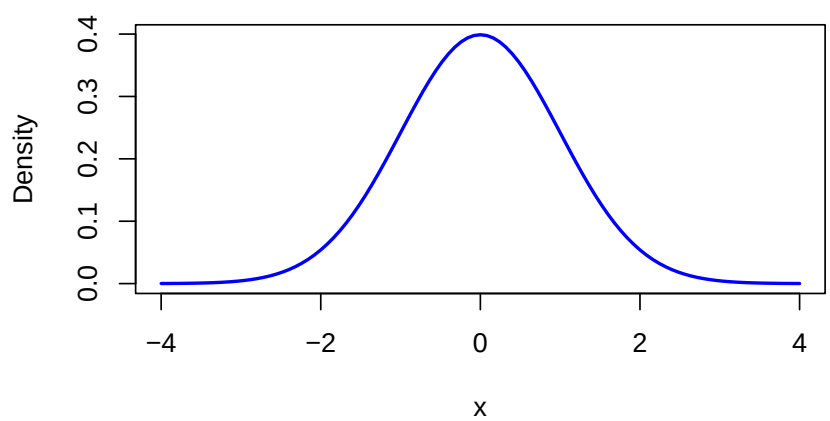
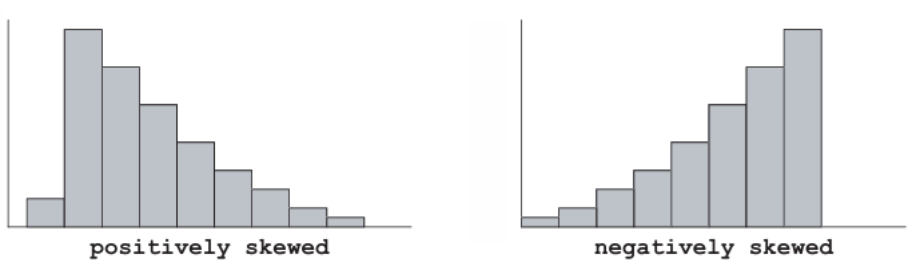
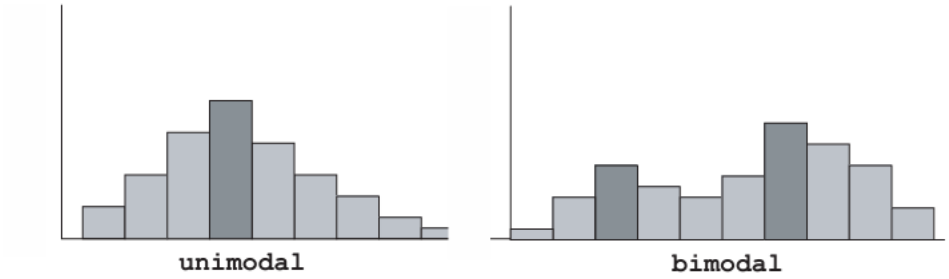


Figure 2.2: Gaussian Distribution

A **skewed histogram** is one with a long tail extending either to the right or to the left:



Histograms can also be **unimodal**, with a single peak, or **bimodal**, with two peaks, and so on.



! Key idea

The *shape* of a histogram provides valuable information about the distribution of the data, beyond what summary statistics alone can reveal.

2.4 Quantitative and qualitative variables

A **variable** is a characteristic that can vary from one individual to another.
For example, if we ask *"How old are you?"*, the variable of interest is **age**.
In this case, age is a *quantitative variable*, since it is usually expressed as a number.

In other cases, the question might be *"What color is your hair?"*
Here, the variable is a *qualitative variable*, because the responses describe categories rather than numerical magnitudes.

A useful distinction among quantitative variables is between **discrete** and **continuous** variables.

- A **discrete variable** takes on values in fixed steps. An example is the number of children in a household.
- A **continuous variable** can take on any value along a real line, such as height, weight, or income.

Working with quantitative variables is usually straightforward. Qualitative variables, however, are often handled by assigning **arbitrary numerical codes** to the possible responses. In practice, this means that a researcher codes qualitative information into nominal numerical data.

For example, a course evaluation might be coded as:

- Poor = 1
- Fair = 2
- Good = 3
- Very Good = 4
- Excellent = 5

> ⚠ Common pitfall
>
> Do not treat coded qualitative variables as numerical measurements. Arithmetic operations on these codes usually have no meaningful interpretation.

It is important to remember that these numbers are **labels rather than true measurements**, and arithmetic operations on them may not be meaningful.

It is also useful to distinguish between different **data structures**, such as:

- **Cross-sectional data**, which observe many individuals at a single point in time
- **Time-series data**, which track a single unit over time
- **Panel (or longitudinal) data**, which follow multiple individuals over time

The statistical tools available to the researcher often depend on which of these data structures is being used.

> ℹ Pause and think
>
> Which of the variables you encounter in daily life—income, grades, or job titles—are quantitative, and which are qualitative?

2.5 Other graphical representations of data

There are many graphical methods available for representing data. Some, such as **bar graphs** and **pie charts**, are widely known. Others, such as the **stem-and-leaf plot** or the **ogive**, are more specialized.

Most introductory statistics textbooks discuss a variety of graphical summaries, and interested readers are encouraged to consult these sources.[3]

! Big picture

Many graphical tools exist, but for understanding distributions and probabilities, the **histogram** plays a uniquely central role.

! Chapter summary

Graphical summaries provide a powerful way to explore and communicate patterns in data.
Different types of data—cardinal, ordinal, and nominal—call for different graphical representations. Among these, the histogram plays a central role in statistics because it summarizes distributions through areas and connects naturally to ideas of probability.
Understanding how to construct and interpret histograms, especially when class intervals are unequal, is essential for sound statistical analysis and for the topics that follow in later chapters.

2.6 Exercises

2.6.1 Conceptual questions

1. Explain why graphical summaries are often more informative than tables when dealing with large data sets.
2. Give one example each of:
   - a cardinal variable,
   - an ordinal variable, and
   - a nominal variable.

   Briefly explain why arithmetic operations make sense for one but not for the others.
3. Why can assigning numbers to qualitative categories be misleading if those numbers are treated as measurements?

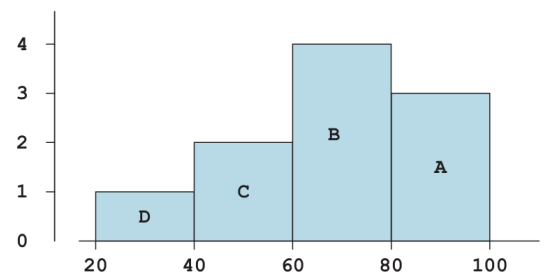
[3] Some recommended texts include Peck, Olsen, and Devore (2005, Chapter 3), Utts and Heckard (2006, Chapter 2), and Keller (2005, Chapters 2 and 3). The exercises at the end of this book also ask for graphical summaries not fully explained here, but which are worth exploring with the help of other introductory texts or online resources.

4. A histogram shows two distributions with the same total area. What does this tell you about the data being represented?

> **Check your understanding**
>
> If two histograms look very different but have the same total area, what does this imply about the percentages or probabilities they represent?

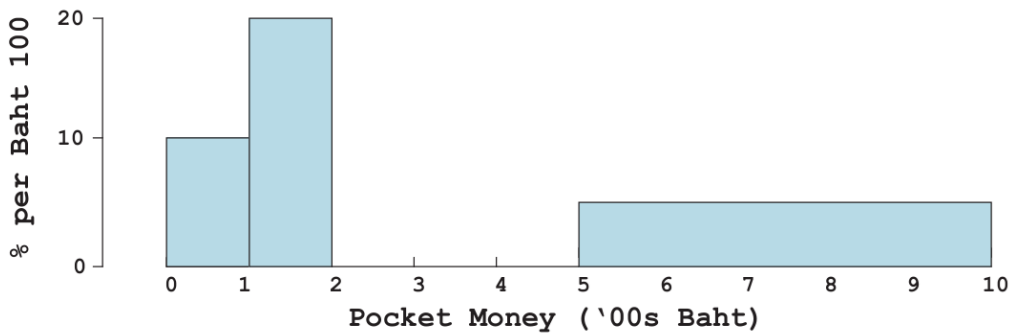
2.6.2 Interpreting histograms

5. Consider a histogram constructed using unequal class intervals.
   - Why must the vertical axis be interpreted as a *density* rather than a raw frequency?
   - What would go wrong if heights were drawn directly from percentages?
6. A histogram is right-skewed.
   - What does this tell you about the distribution of the data?
   - Give a real-world example of a variable that is likely to be right-skewed.
7. Explain the difference between a unimodal and a bimodal histogram.
   What might cause a bimodal shape in real data?
8. The histogram below shows the distribution of final scores in a certain class.

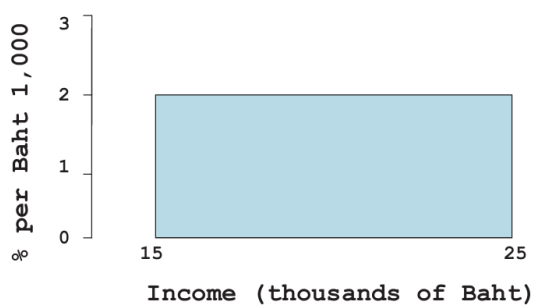


   - which block represents the people who scored between 60 and 80?
   - Ten percent scored between 20 and 40. About what percentage scored between 40 and 60?
   - About what percentage scored over 60?
   9.A histogram of pocket-money used by EBA students in a week is shown above. No student used over Baht1,000 per month. The block over the class interval 200-500

is missing. How tall must it be?



9. The figure below shows a block of family income in a certain town. About what percent of the families in the city had incomes between Baht 15,000 and 25,000?



> ⚠ Common pitfall
>
> Students often confuse the *height* of bars with the *area* of bars in a histogram.

2.6.3 Data structure and graphs

11. For each of the following data structures, suggest an appropriate graphical summary and explain why: - cross-sectional data, - time-series data, - panel (longitudinal) data.
12. Why is the histogram particularly important for later topics in probability and statistical inference?

2.6.4 Optional challenge

13. Find a real dataset (from news, government statistics, or online sources) and:
    - classify the variables as cardinal, ordinal, or nominal,
    - propose at least one appropriate graphical summary for each variable,
    - explain why a histogram would or would not be appropriate.

3 Numerical Summaries

> **Learning objectives**
>
> By the end of this chapter, you should be able to:
>
> - explain the purpose of numerical summaries in descriptive statistics,
> - distinguish between different measures of central location,
> - describe how dispersion captures variability in data,
> - compute and interpret the mean, median, and standard deviation,
> - explain the difference between population and sample standard deviation,
> - interpret percentiles and quartiles as measures of relative standing.

Descriptive statistics consist not only of graphical summaries, but also of **numerical methods**.
Since we introduced the histogram in the previous chapter—the most important graphical summary of data—we now turn to **basic numerical descriptive statistics**.

These include:

- **Measures of central location**, such as the mean, median, and mode
- **Measures of dispersion (or spread)**, such as the range, variance, standard deviation, and coefficient of variation
- **Measures of relative standing**, such as percentiles and quartiles

Together, these numerical summaries help us describe data in a concise and informative way.

> **Big picture**
>
> Numerical summaries allow us to describe large datasets using just a few carefully chosen numbers.

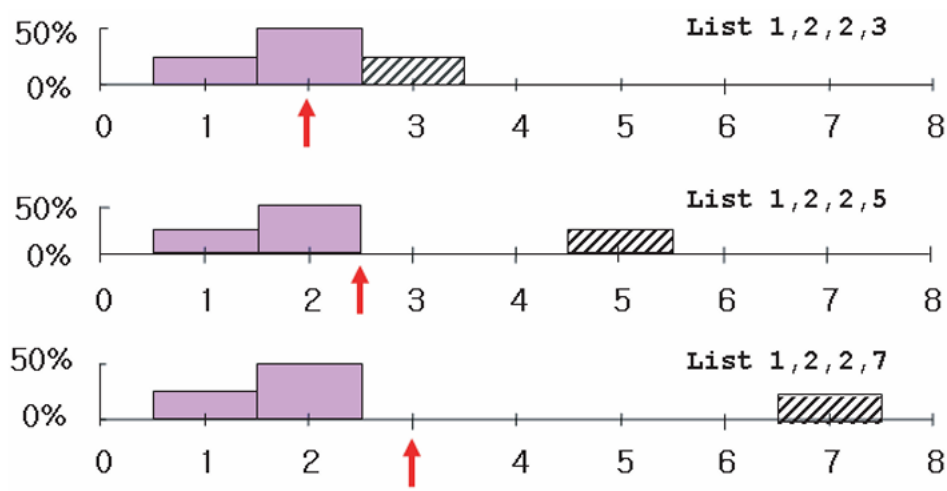
3.1 Measure of central location

The **arithmetic mean** (or simply, the *average*) is the most widely used measure of central location. It is calculated by adding up all observed values and dividing by the number of observations.

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{3.1}$$

The arithmetic mean, shown formally as Equation 3.1, is **unique**. Moreover, it has an important and intuitive property: the sum of the deviations of each observation from the mean is zero.

An interesting way to visualize this property is through a histogram. If the histogram were placed on a balance, it would balance exactly when supported at the arithmetic mean.



The **median**, another measure of central location, is the midpoint of the values in a dataset once the observations are ordered from the smallest to the largest.

The **mode** is a less commonly used measure of central location and refers to the value that occurs most frequently in the data.

An important advantage of the median is that it is **not affected by extremely large or small values**. For this reason, it is often a more reliable measure of central location when such extreme values are present.

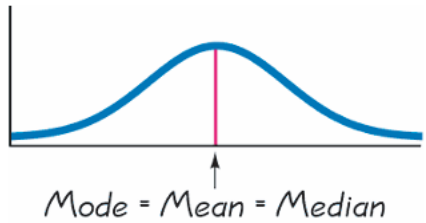
The mode, on the other hand, can be interpreted as the *"fashion category"* of the data. In a histogram, it corresponds to the tallest bar or the highest peak.

Key idea

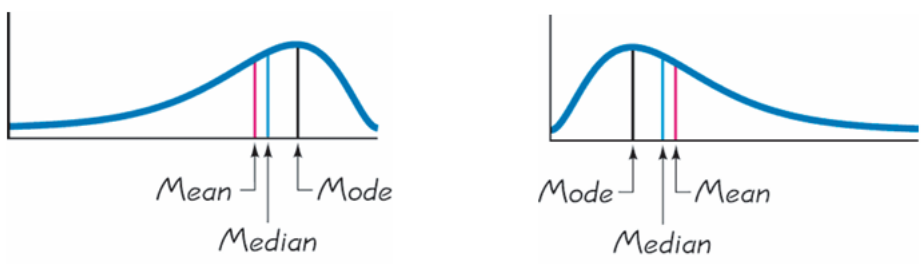
The mean, median, and mode summarize the *center* of the data, but they respond differently to extreme values.

For **symmetrical histograms**, such as the Gaussian (normal) distribution, the three measures coincide:

$$\text{mean} = \text{median} = \text{mode}.$$



But not all histograms are symmetric. If a histogram is skewed to the left or to the right, the three measures may differ.

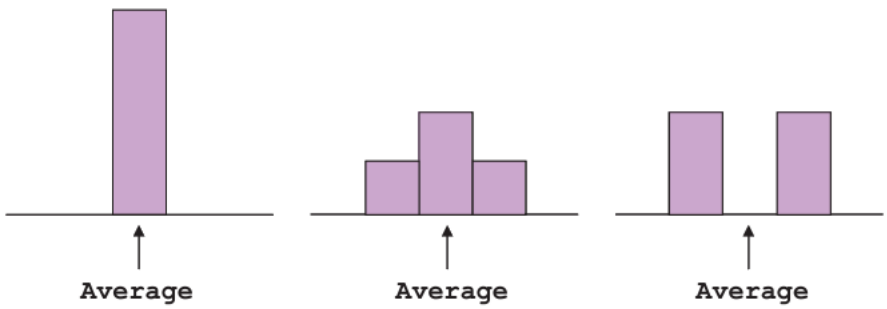


What is important to realize is that **extreme values affect the average most, then the median, and least the mode**. The arithmetic mean is therefore the most sensitive to extremely large or small values.

Common pitfall

Relying only on the mean can be misleading when the data contain extreme values or are highly skewed.

3.2 Measures of dispersion or spread



The three histograms above all have the same average (or mean), yet they clearly look different. Describing a distribution using only its mean is therefore not sufficient. We also need to know how **spread out** the data are—that is, the **dispersion** or **variability** of the observations.

There are many measures of dispersion, including the range, the mean absolute deviation, the root mean square (r.m.s.), the standard deviation, the variance, the coefficient of variation, and the interquartile range. We now focus on the most important of these.

3.2.1 The standard deviation

The **standard deviation**, abbreviated as **SD**, is the most important measure of dispersion used in this book. There are several ways to define and compute it, but the most intuitive approach is based on the idea of the **root mean square (r.m.s.)**.

The starting point is the notion of a **deviation**, which is simply the distance of an observation from the average. Mathematically, the deviation of an observation (X_i) from the mean ($\overline{X}$) is

$$\text{deviation} = X_i - \overline{X}.$$

The term **root mean square** is a useful reminder of the steps involved—provided one reads it backwards:

1. **Square** all the entries

2. Take the **mean** of the squares

3. Take the square **root** of the mean

In equation form, the root mean square of a variable X can be written as

$$\text{r.m.s. of } X = \sqrt{X^2}.$$

This, however, is just a *mathematical operation*.

The **standard deviation** is defined more specifically as the root mean square of the **deviations from the mean**, that is,

$$\text{SD of } X = \text{r.m.s. (deviation from the average)}.$$

In simple terms, the standard deviation tells us how far, on average, the observations lie from their mean.

3.2.2 A numerical example

To make this concrete, consider the list of numbers

$$20,\ 10,\ 15,\ 15.$$

The average $\overline{X}$ is 15. The deviations from the mean are therefore

$$5,\ -5,\ 0,\ 0.$$

Applying the r.m.s. procedure gives the standard deviation:

$$\begin{aligned}
\text{SD} &= \sqrt{\frac{5^2 + (-5)^2 + 0^2 + 0^2}{4}} \\
&= \sqrt{\frac{25 + 25 + 0 + 0}{4}} \\
&= \sqrt{\frac{50}{4}} = \sqrt{12.5} \approx 3.5.
\end{aligned}$$

Try This: Exploring Mean and SD

Part 1: Shifting and Scaling
Find the mean and standard deviation (SD) for the following lists:
(i) $1, 3, 4, 5, 7$ **(ii)** $6, 8, 9, 10, 12$ **(iii)** $3, 9, 12, 15, 21$
Question: What do you notice about how the mean and SD change across these three lists?

Part 2: Symmetry and Signs
Find the mean and SD for these two lists:
(i) $5, -4, 3, -1, 7$ **(ii)** $-5, 4, -3, 1, -7$

- Adding a constant to a list does not change the standard deviation.
- Multiplying a list by some constant k multiplies the standard deviation by k
- Changing each sign of elements in a list does not change teh standard deviation

3.2.3 Population and sample standard deviation

It is important to distinguish between the **population standard deviation** and the **sample standard deviation**, although this distinction becomes more relevant in statistical inference.

When computing the population standard deviation, we divide by n, the number of observations:

$$\text{SD} = \sqrt{\frac{\sum(X - \overline{X})^2}{n}} \tag{3.2}$$

When computing the standard deviation of a **sample**, we divide by $(n - 1)$:

$$\text{SD}^+ = \sqrt{\frac{\sum(X - \overline{X})^2}{n - 1}} \tag{3.3}$$

We use SD^+ to denote the sample standard deviation. It is numerically larger than SD because we divide by the smaller number $(n - 1)$ rather than n.[1]

3.2.4 Variance and the coefficient of variation

Another measure of dispersion is the **variance**, which is simply the square of the standard deviation. While the standard deviation is measured in the same units as the data, the variance is measured in squared units, making interpretation less direct.[2]

[1] This is for notional purposes only. Although it may seem counterintuitive, this adjustment corrects for the fact that samples tend to underestimate population variability.

[2] Variances are especially useful in algebraic manipulations. For example, variances can be added, whereas standard deviations cannot.

Comparing variability across variables measured in different units is often difficult. A **relative measure of dispersion**, known as the **coefficient of variation (CV)**, is sometimes used. The CV is defined as the ratio of the standard deviation to the mean, expressed as a percentage.

3.3 Measures of relative standing

Many introductory statistics textbooks also discuss **measures of relative standing**, such as **percentiles** and **quartiles**, as well as graphical summaries like the **box plot**, shown below.[3]

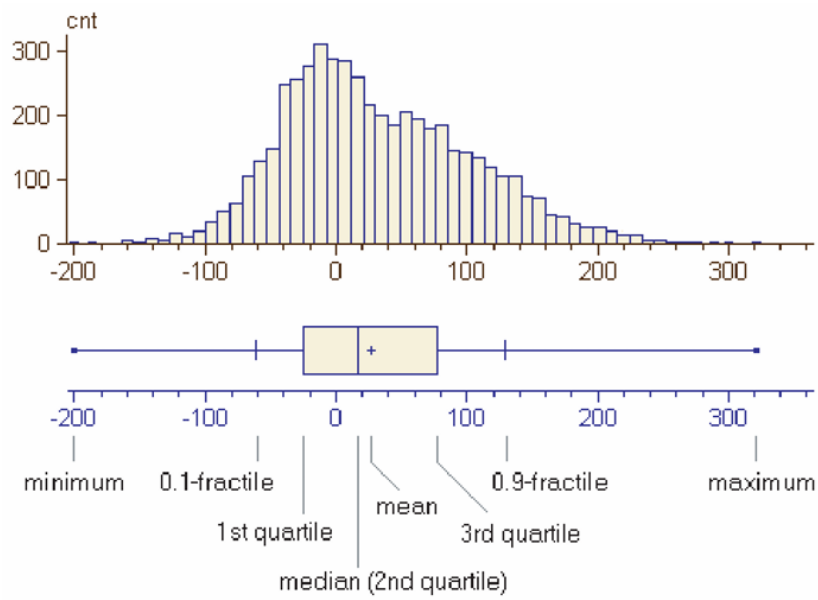


Figure 3.1: A box-plot example

A **percentile** refers to the percentage of observations lying below a given value.

A unique maximum corresponds to the **100th percentile**, while a unique minimum corresponds to the **0th percentile**.

It follows that the **median** is the **50th percentile**. The **25th percentile** and **75th percentile** are known as the **first quartile** and **third quartile**, respectively.

[3] Additional examples are provided in standard textbooks and in the PowerPoint slides accompanying this book.

Key idea

Measures of relative standing tell us *where* an observation lies within a distribution, not just its magnitude.

Chapter summary

Numerical summaries allow us to describe key features of a dataset using a small number of carefully chosen statistics.
Measures of **central location** describe where the data are centered, while measures of **dispersion** describe how spread out the observations are. Measures of **relative standing** indicate how individual observations compare with the rest of the data.
Together, these tools form the foundation for understanding data and prepare us for statistical inference in later chapters.

3.4 Exercises

3.4.1 Conceptual questions

1. Explain why the mean alone is often insufficient to describe a dataset.
2. Compare the mean, median, and mode.
   In what situations might one be preferred over the others?
3. Why is the median often a better measure of central location than the mean when data contain extreme values?
4. Explain why two datasets can have the same mean but very different standard deviations.

Check your understanding

If two datasets have the same mean and median, must they have the same standard deviation? Explain.

3.4.2 Understanding dispersion

5. Describe in words what the standard deviation measures.
   Why is it based on deviations from the mean rather than deviations from the median?
6. Explain the role of the root mean square (r.m.s.) in defining the standard deviation.
7. Which has larger r.m.s? the list $7, 7, 7, 7$ or $7, -7, 7, -7$?
8. What is the r.m.s of $17, 17, 17, 17, 17$? What is the SD?
9. Can the SD ever be negative?
10. For a list of positive numbers, can the SD ever be larger than the average?
11. Why does the variance have squared units, and why can this make interpretation difficult?

> ⚠ Common pitfall
>
> A small standard deviation does not imply that the data are "small"—only that they are tightly clustered around the mean.

3.4.3 Measures of relative standing

12. Explain what it means for a value to be at the 75th percentile of a distribution.
13. Why is the median equal to the 50th percentile?
14. Describe how a box plot summarizes information about central location, dispersion, and relative standing.

3.4.4 Numerical practice

15. Consider the dataset:

$$5,\ 8,\ 2,\ 9,\ 5,\ 3,\ 7,\ 4,\ 2,\ 7,\ 4,\ 10,\ 4,\ 3,\ 5.$$

a. Compute the mean, median, and mode.

b. Compute the standard deviation, SD and variance.

c. Explain what each statistic tells you about the data.

d. Calculate their 25th and 75th percentiles.

e. Find the inter-quartile range (IQR).

f. Calculate the coefficient of variation for the two lists.

g. What is the mean absolute deviation for both lists?

h. Draw a boxplot for the lists of numbers.

3.4.5 Optional challenge

16. Find a real dataset (for example, income, test scores, or prices):

- compute at least two measures of central location,
- compute at least one measure of dispersion,
- explain which statistics are most informative and why.

4 Correlation

Learning objectives

By the end of this chapter, you should be able to:

- distinguish between univariate and bivariate data,
- use graphical tools to describe relationships between two variables,
- explain why means and standard deviations are insufficient for describing association,
- define and interpret the correlation coefficient,
- compute correlation using standard units,
- understand the limitations of correlation, especially with respect to causation.

4.1 Graphical representation

So far, we have focused on a single variable—what is known as **univariate data**—and examined their graphical and numerical summaries. We now turn to **bivariate data**, that is, data involving two variables.

As economists, we are often interested in understanding relationships between variables—for example, wages and education, CEO performance and salaries, economic growth and foreign aid, and many others. As with univariate data, both **graphical** and **numerical** summaries can be used.

Key graphical tools for two variables

The two most important graphical summaries for bivariate data are:

- **Contingency tables**, usually for categorical variables
- **Scatter diagrams**, usually for quantitative variables

A **contingency table** lists the frequency of each combination of values of two categorical variables. For example, a survey of 2,237 people recording gender and handedness might

produce the following table.[1]

Table 4.1: Gender and handedness

	Male	Female
Right-handed	934	1070
Left-handed	113	92
Ambidextrous	20	8

When dealing with **cardinal (quantitative) data**, the relationship between two variables is better visualized using a **scatter diagram**. The figure below shows the association between the duration of eruptions of Old Faithful and the waiting time between eruptions.

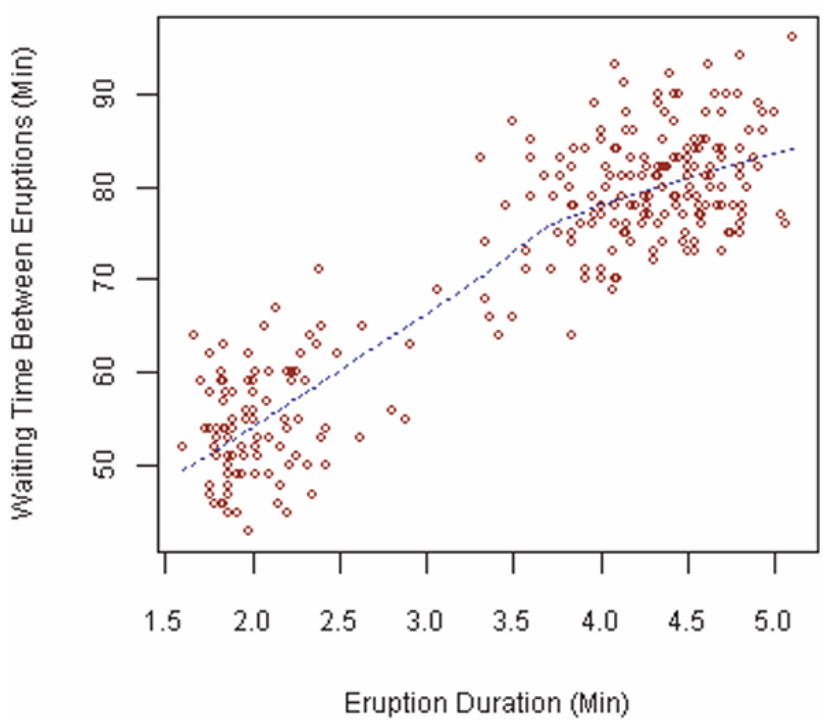


Figure 4.1: Waiting time and eruption duration of Old Faithful

4.2 Correlation

Graphs are informative, but they also have limitations. For this reason, we need **numerical summaries** that describe the relationship between two variables.

[1]For more on the relationship between categorical variables, a useful text is Utts and Heckard (2006, Chapter6).

In earlier chapters, we introduced the **mean** and **standard deviation** as the most important numerical summaries for univariate data. One might therefore ask: *Why are these not sufficient for bivariate data?*

> **!** Why means and SDs are not enough
>
> Two datasets can have the same mean and standard deviation for each variable, yet exhibit very different relationships between those variables.

Consider a university that has offered two statistics sections, A and B, over several years. For each student, midterm and final exam scores (out of 200) are recorded and plotted.

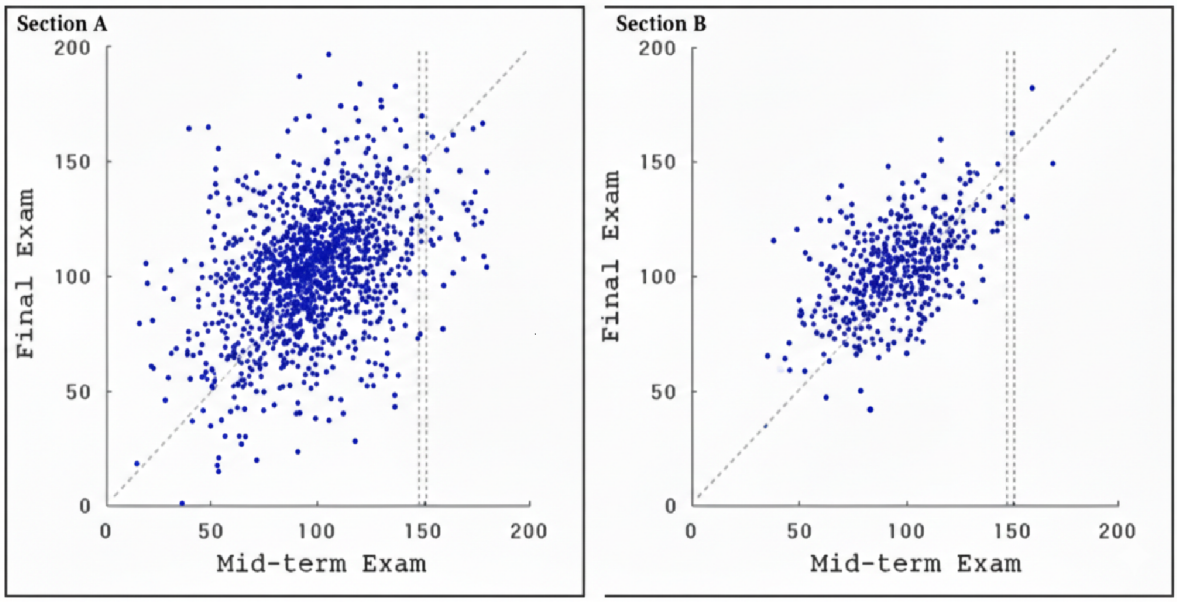


Figure 4.2: Students' score: Section A and B

Points lying on the 45° line represent students whose midterm and final scores are the same.

If there is a **strong association**, knowing one variable helps predict the other. If the association is weak, knowing one variable provides little predictive power.

For example, a student in section A who scored 150 on the midterm could plausibly score anywhere between 60 and 160 on the final. In section B, a student with the same midterm score would likely score between 100 and 160. Midterm scores therefore provide better predictive information in section B.

More strikingly, the **means** and **standard deviations** of both midterm and final scores are almost identical across the two sections. This means that we cannot distinguish the sections using only these summaries.

! We need a new measure

To capture how tightly points cluster around a line in a scatter diagram, we introduce the **correlation coefficient**, denoted by r.

4.3 Correlation coefficient

The **correlation coefficient** is a pure number between -1 and 1.

! Important

$$-1 \leq Corr(X, Y) \leq 1$$

Furthermore:

- $r = 1$ or $r = -1$ indicates **perfect correlation**
- A positive value of r indicates a **positive relationship**
- A negative value of r indicates a **negative relationship**

The figures below show two variables with the same mean (3) and standard deviation (1), but different correlations.

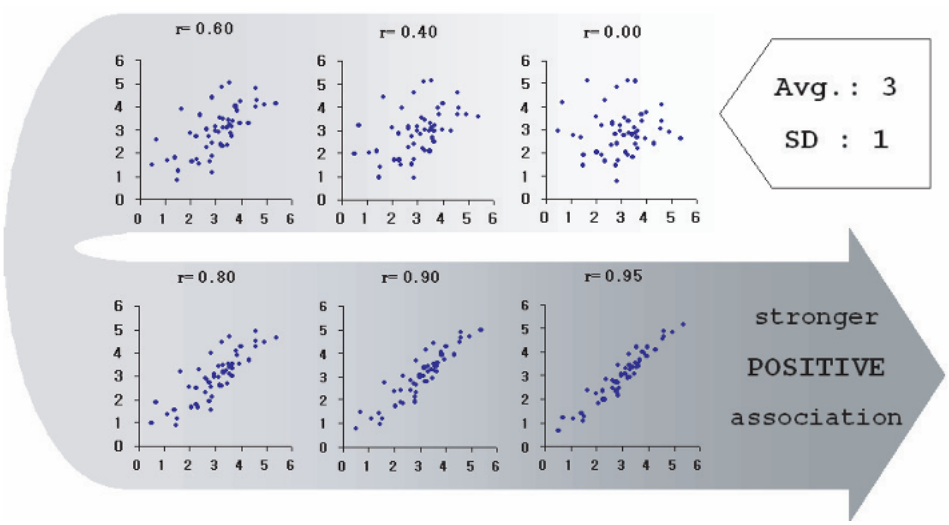


Figure 4.3: Positive correlation

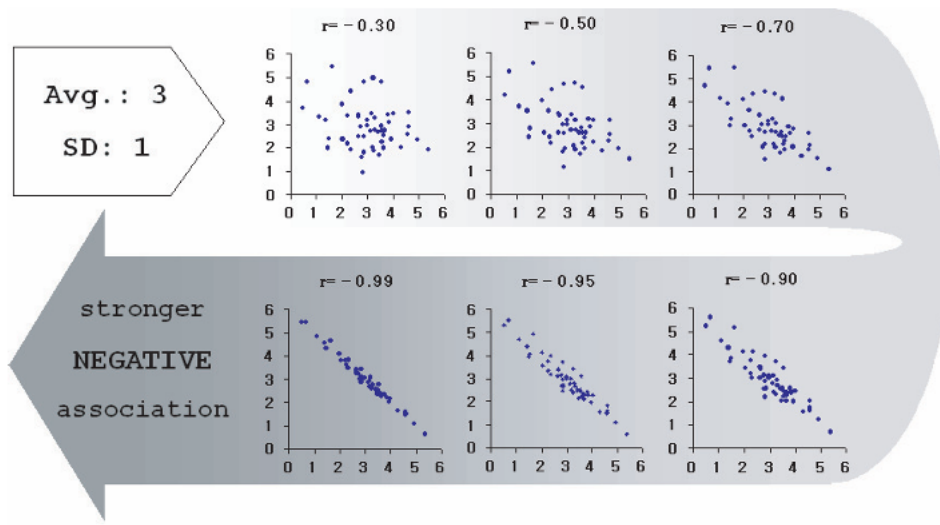


Figure 4.4: Negative correlation

> **A technical detail**
>
> If either variable has a standard deviation equal to zero, then $r = 0$. Correlation requires variation in *both* variables.

4.3.1 How correlation is computed

To compute the correlation coefficient:

1. Convert each variable into **standard units**
2. Multiply the standard units pairwise
3. Take the **average** of these products

Standard units measure distance from the mean in terms of standard deviations.

For the list $1, 3, 4, 5, 7$, the mean is 4 and the standard deviation is 2. The values in standard units are: $-1.5,\ -0.5,\ 0,\ 0.5,\ 1.5$

> **Standard units**
>
> Standard units measure how far a value lies from the mean relative to the spread of the data.

Now consider the five points: $(1, 5), (3, 9), (4, 7), (5, 1), (7, 13)$.

The mean of X is 4 with $SD_X = 2$, and the mean of Y is 7 with $SD_Y = 4$.

Table 4.2: Calculating the correlation coefficient

X	Y	SU_X	SU_Y	Product
1	5	-1.5	-0.5	0.75
3	9	-0.5	0.5	-0.25
4	7	0.0	0.0	0.0
5	1	0.5	-1.5	-0.75
7	13	1.5	1.5	2.25

The correlation coefficient is the **average of the last column**, which equals $r = 0.4$.

4.3.2 Why this method matters

This approach shows that correlation:

- has **no units**
- is unchanged by rescaling variables
- measures **relative**, not absolute, clustering

There are many other ways to compute the correlation coefficient, but we will stick to the above method as it reveals best how the correlation coefficient works.

In many textbooks, the formula for correlation is written as:

! Correlation coefficient

$$Corr(X, Y) = \frac{\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \overline{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \overline{Y})^2}}$$

Another definition commonly used is:

$$Corr(X, Y) = \frac{\text{Covariance}(X, Y)}{SD_x SD_y}$$

Exercise: Correlation coefficients

Compute the **correlation coefficient** for each of the following datasets.
(a)

X	1	1	1	1	2	2	2	3	3	4
Y	5	3	5	7	3	3	1	1	1	1

(b)

X	1	1	1	1	2	2	2	3	3	4
Y	1	2	1	3	1	4	1	2	2	3

(c)

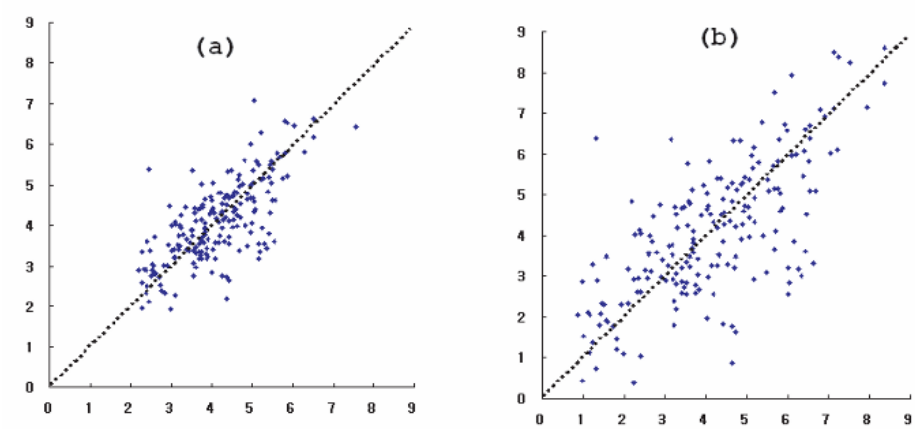
X	1	1	1	1	2	2	2	3	3	4
Y	2	2	2	2	4	4	4	6	6	8

Question.
What do you observe about the values of the correlation coefficients across the three cases?

Note that the correlation coefficient is a pure number without any units, which is noted by the conversion to standard units.

Moreover, the correlation coefficient is *not** affected by any of the three changes: - 1) interchanging the two variables, - 2) adding the same number to all the values of one variable, and - 3) multiplying all the values of one variable by the same positive number.

This basically means that changing the scale of any of the variables leaves the correlation coefficient unchanged.



Be warned, looks can be deceiving. A quick glance at the above scatter plots would easily convince someone that the points in (a) is more tightly clustered, and thus having a correlation coefficient closer to 1, than (b). This cannot be further than the truth, because in fact = 0.7 for both diagrams. (b) is simply a magnification of (a). What is important to realize is that calculating involves converting variables to standard units (deviation from average divided by SD) and therefore it measures clustering not in absolute terms, but in relative terms (more specifically, relative to their standard deviations). That is, although plot (a) may appear more tightly clustered than plot (b), both have $r = 0.7$. Plot (b) is simply a magnified version of plot (a). Correlation measures clustering **relative to standard deviation**, not in absolute terms.

> ⚠ Correlation measures linear association
>
> The correlation coefficient captures **linear relationships only**. Nonlinear relationships may have low correlation even when variables are strongly related.

4.4 Association is not causation

The last piece of warning when using correlation coefficients is to realize that what we have is a representation of association, that is how the variables move relative to each other. The correlation coefficient does not establish causation. This is best left to economic theory, which is called upon to interpret and explain the data.

> ❗ Important
>
> Correlation measures **association**, not **causation**.

A positive or negative correlation does not explain *why* variables move together. Establishing causation requires economic theory and careful empirical reasoning.

For example, education is often thought to cause higher wages because more educated individuals tend to earn more. However, a third factor—such as **individual determination or motivation**—may influence both education and wages.

Such a factor is known as a **confounding variable**, and its presence can invalidate causal interpretations based solely on correlation.

> ❗ Takeaway
>
> Correlation is a powerful descriptive tool, but causal claims require much more than a high value of r.

Chapter summary

Correlation provides a numerical summary of the strength and direction of linear association between two variables.
Scatter diagrams help visualize relationships, while the correlation coefficient measures how tightly points cluster around a straight line. However, correlation alone cannot establish causation, and careful reasoning is required to interpret empirical relationships correctly.

4.5 Exercises

4.5.1 Conceptual questions

1. Explain the difference between univariate and bivariate data.
2. Why are scatter diagrams useful for studying relationships between quantitative variables?
3. Explain why two datasets can have identical means and standard deviations but very different correlations.

4.5.2 Understanding correlation

4. What does the sign of the correlation coefficient indicate?
5. What does the magnitude of the correlation coefficient tell us?
6. Why must both variables exhibit variation for correlation to be meaningful?

Common pitfall

A high correlation does not imply a causal relationship.

4.5.3 Computation and interpretation

7. Explain how converting variables to standard units removes units from the correlation coefficient.
8. Why is correlation unaffected by changes in scale or units?
9. Give an example of a nonlinear relationship that might have low correlation.

4.5.4 Optional challenge

10. Find two real-world variables that are correlated.
    Discuss whether the relationship is likely causal, and identify possible confounding variables.

5 Elementary Probability

Learning objectives

By the end of this chapter, you should be able to:

- explain the main interpretations of probability,
- define experiments, outcomes, sample spaces, and events,
- compute conditional probabilities,
- apply the multiplication and addition rules,
- distinguish between independence and dependence,
- understand concept of mutually exclusive,
- use Bayes' rule to update probabilities,
- understand the axioms underlying probability theory.

In this chapter, we introduce the basic ideas of **probability** needed to understand statistical concepts and reasoning. The literature typically identifies **three definitions of probability**:

1. the **classical** definition,

2. the **empirical (frequentist)** definition, and

3. the **subjective** definition.

The **classical definition** assumes that all outcomes of an experiment are equally likely. For example, when rolling a fair die, the probability of obtaining an even number is the ratio of favorable outcomes (2, 4, 6) to total possible outcomes (1–6), which is $3/6 = 1/2$.

The **empirical (frequentist) definition** is based on observed relative frequencies. For example, one might talk about the probability that Manchester United beats Chelsea, based on the fraction of past games won by Manchester United.

The **subjective definition** reflects an individual's beliefs, informed by experience and available information. For instance, a student may assign a subjective probability to the event that they will earn an A in a statistics course like this one.

A unifying point

Although these interpretations differ philosophically, the **mathematics of probability** that follows does not depend on which definition we adopt.

5.1 The probability space, experiment, and outcome

Consider tossing a coin. The outcome is uncertain: it may land **heads** or **tails**. In probability language, tossing the coin is an **experiment**, and the result is an **outcome**.

An experiment produces exactly **one outcome**, chosen from a set of possible outcomes. This set is called the **sample space**, denoted $\mathcal{S}$. A **subset** of the sample space is called an **event**.

There is no restriction on what constitutes an experiment. Tossing one coin once, tossing it three times, or even tossing it infinitely many times can each be considered a single experiment.

To make this concrete, consider tossing a coin **three times**. The sample space is

$\mathcal{S} = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$.

If we assume each outcome is equally likely, then each has probability 1/8.

A basic fact

Probabilities always lie between 0 and 1 (or 0% and 100%).
More formally:

$$0 \leq Prob(A) \leq 1 \tag{5.1}$$

From this, it follows that for any event A,

$$P(\sim A) = 1 - P(A) \tag{5.2}$$

where $\sim A$ denotes the complement of A or (not A).

5.2 Conditional probability

Conditional probability is best understood through an example.

Suppose a deck of cards is shuffled and two cards are drawn. You win 100 baht if the **second card** is the ACE of hearts. Before seeing any cards, the probability of winning is 1/52.

Now imagine that the **first card is revealed** and it is the seven of clubs. What is your probability of winning now?

Since one card is no longer in the deck (and you know what it is), the probability becomes 1/51.

Put differentlay, when pulling out two cards, because you now know the first card, the probability of the second card is a conditional probability i.e. the probability of a particular event occurring, given that another event has occurred. This is compactly written as $P(A \mid B)$.

! Conditional probability

Conditional probability reflects how probabilities change **once new information becomes available**.

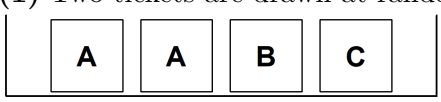
Formally, the probability that event A occurs **given** that event B has occurred is written

$$P(A \mid B) \tag{5.3}$$

which is read as the probability of A given B.

Try This: Conditional Probability

(1) Two tickets are drawn at random without replacement from the following box:



(a) What is the chance that the second ticket is $\boxed{\text{B}}$?
(b) What is the chance that the second ticket is $\boxed{\text{B}}$, given thayt the fist is $\boxed{\text{A}}$?

(2) Three cards are drawn from a deck of cards without replacement. What is the probability that none of them will be hearts

5.3 The multiplication rule

Consider a box containing three tickets labeled R, W, and B. Two tickets are drawn **without replacement**. What is the probability of drawing R first and then W?

On the first draw, the probability of R is 1/3. Given that R was drawn, only W and B remain, so the probability of W on the second draw is 1/2. Therefore,

$P(R \text{ then } W) = \frac{1}{2} \text{ of } \frac{1}{3} = \frac{1}{6}.$

This illustrates the **multiplication rule**:

$$P(AB) = P(B \mid A)P(A) = P(A \mid B)P(B) \tag{5.4}$$

For example, the probability that **two cards dealt from a deck are both ACES** is $\frac{3}{51} \times \frac{4}{54}$.

5.4 Independence

Two events are said to be **independent** if knowing the outcome of one does not affect the probability of the other.

Drawing with replacement produces independent events; drawing without replacement produces dependent events.

! Independence

Events A and B as said to be **independent** if

$$P(A \mid B) = P(A) \tag{5.5}$$

(or equivalently $P(B \mid A) = P(B)$).

When we have independence, the multiplication rule conveniently simplifies to

$$P(AB) = P(A)P(B) \tag{5.6}$$

Try these: (1) A fair coin is tossed twice. What is the chance of getting a head then a tail? **(2)** A die is rolled three times. (i) Find the chance that the first roll is an ACE. (ii) Find the chance that the first roll is an ACE, the second roll is a deuce, and the third roll is a trey..

5.5 Checking for independence

Suppose we want to know whether graduating from GA program helps someone become a successful manager/CEO.

Let:

- A_i: GA graduate `(=1)`, or 0 otherwise

- B_i: Becomes CEO `(=1)`, or 0 otherwise

The table below summarizes survey results:

	Becomes CEO	Does not become CEO	**Total**
GA graduate	0.11	0.29	0.40
Not GA graduate	0.06	0.54	0.60
Total	0.17	0.83	1.00

The **marginal probability** of becoming a CEO is 0.17. The **joint probability** of a GA graduate becoming a CEO is 0.11, and so on.

! Marginal probability from joint probabilities

To get the *marginal* probability of a student being a GA graduate, we add the two joint probabilities of the first row $(0.11 + 029)$ to get 0.4. Note here the two jint probabilities that we add, the probability that we have a GA graduate and CEO *plus* the probability that we have a GA graduate and not a CEO. Here probabilty of GA is the same, but we have CEO and not CEO. So by adding across CEO type, we essentially **integrate it out** and we are left with the marginal probabilty of 0.4, the probability of GA graduate.

To test independence, compute

$$P(A_1 \mid B_1) = \frac{P(A_1 B_1)}{P(B_1)} = \frac{0.11}{0.17} \approx 0.65.$$

Since $P(A_1 \mid B_1) \neq P(A_1)$, the two events are **not independent**.

Interpretation

Becoming a CEO is related to being a GA graduate in this example.

5.6 The addition rule

Two events are said to be **mutually exclusive** if the occurrence of one prevents the occurrence of the other.

For mutually exclusive events,

$$P(A \text{ or } B) = P(A) + P(B) \tag{5.7}$$

For example, intuitively, if we draw one card from a deck of cards, the probability of getting either a heart or spades is $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$. The additional rule tells us to simply add the chances provided that the two events are mutually exclusive. The probability to get a heart is 1/4 and the probability to get a spade is also 1/4. Are the two events mutually exclusive? Yes, getting a heart means you can't get a spade, and vice versa. So simply add the chances to get $1/4 + 1/4 = 1/2$.

However, for events that are **not mutually exclusive**, the general addition rule is

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) \tag{5.8}$$

For example, if we toss two dice at the same time, to get at least ace on the two dice, we calculate the probability using the general additional rule, i.e we add the chances and subtract the joint probability because the two events are not mutually exclusive since having an ace on one of the die does not exclude the possibility to get an ace on the other die. In other words, it is possible to get ACE in both dice, i.e. the events are **not** mutually exclusive! Hence we get $1/6 + 1/6 - 1/36 = 11/36$.

! Mutually exclusive events

Events A and B as said to be **mutually exclusive** if

$$P(A \text{ and } B) = 0 \tag{5.9}$$

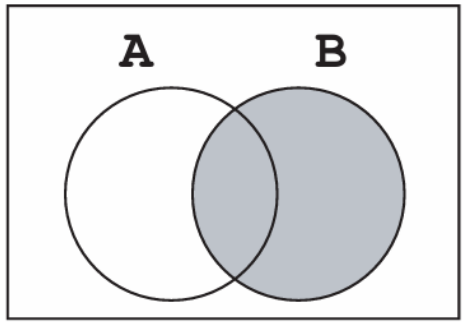
Note that we have used the word 'or' and 'at least' for the additional rule, which can be contrasted with 'and' for the multiplication rule. In terms of set operations, these are the union and intersect operators, respectively.

! Try these.

(1) A die is rolled twice. What is (a) the chance that the first roll is an ace or the second roll is an ace? (b) the chance that the first roll is an ace and the second roll is an ace? (c) the cace of getting at least one ace?
(2) A deck of cards is shuffled. What is (a) the chance that the top card is an ace of spades or the bottom card is an ace of spades? (b) the chance that the first card is an ace of spades and the bottom card is an ace of spades?

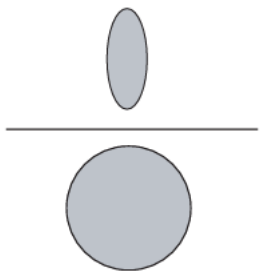
5.7 Venn diagrams

Venn diagrams are very useful when thinking about simple probability problems.
For example, two events A and B that are not disjoint can be drawn as follows:



Even more useful, the conditional probability can be shown by the following:

$P(A|B) =$

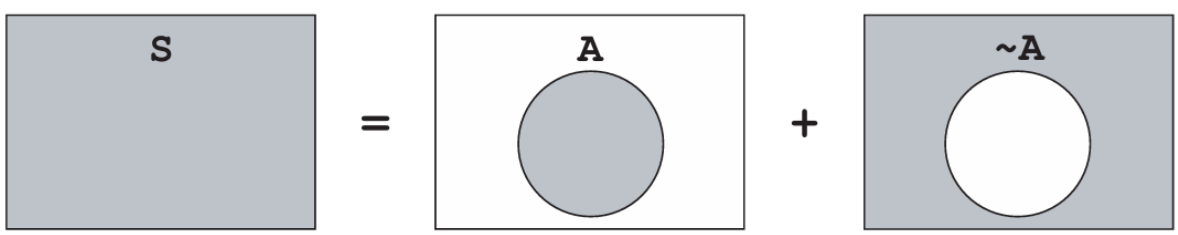


or $\frac{P(AB)}{P(B)}$. That is, given B we wish to know A.

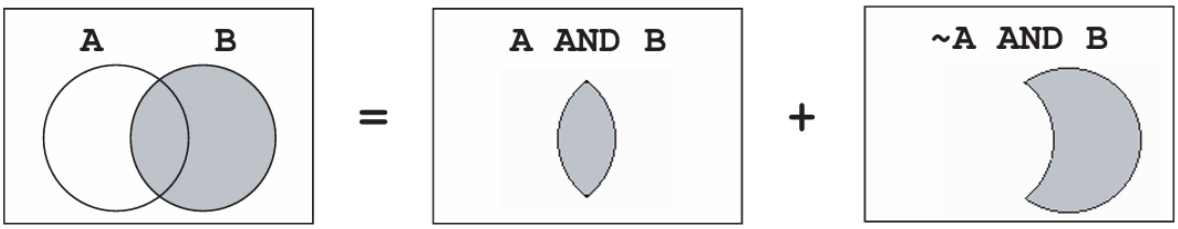
5.8 Bayes' rule

We can easily derive the Bayes' Rule using Venn diagrams.

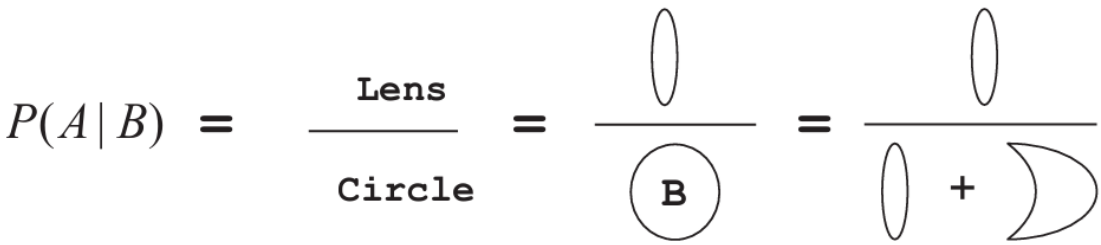
Taking a sample space with event A:



Similarly, the probability of B can be shown using the Venn diagram as follows: \



Combining the two Venn diagrams we get:\



Or what is the same thing

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{P(A \text{ and } B)}{P(A \text{ and } B) + P(\sim A \text{ and } B)}$$

And finally from the definition of joint probabilities we have the Bayes Rule.

! Bayes' Rule

$$P(A \mid B) = \frac{P(A)P(B \mid A)}{P(A)P(B \mid A) + P(\sim A)P(B \mid \sim A)} \tag{5.10}$$

So how do we use the Bayes' Rule? An example should illustrate this. Assume that on a dark night, there was a hit and run accident. A witness, who says that she saw a blue taxi, agrees to sit in the court to testify for the victim so that the latter might get some compensation from the blue taxi company. Incidentally, there are 200 taxis in this town; 170 or 85 % belong to the black taxi company and 30 or 15 % are blue. According to tests conducted under the same condition, the witness identifies blue taxi about 80 % of the time. Now, the question is, What is the probability that she was right, that she really did see a blue taxi that night?

- 15% of taxis are blue
- 85% are black
- The witness correctly identifies blue taxis 80% of the time

- She incorrectly identifies black taxis as blue 20% of the time

To use Bayes' formula, all we have to identify are a few marginal and conditional probabilities; $P(A)$ is the probability that the taxi involved in the accident is blue (i.e 15 %.) Hence $P(\sim A)$ that is the probability of a black taxi, is 85 %. $P(B \mid A)$ is the probability that the witness claims to have seen blue when a blue taxi was really involved, in other words, her accuracy in identifying a blue taxi correctly under the conditions. Conversely, the only other probability we need now to complete the equation is $P(B \mid\sim A)$, which is the probability that the witness sees a blue taxi when in fact a black taxi was involved. i.e 20 %. Plugging all the relevant probabilities gives:

$$P(A \mid B) = \frac{0.15 \times 0.80}{(0.15 \times 0.80) + (0.85 \times 0.20)} = \frac{0.12}{0.29} \approx 0.41.$$

The probability that the witness says she saw "blue" and that it was really a blue taxi is only 0.41, hardly sufficient for the courts to ask the blue taxi company to pay compensation![1]

A sobering result

Even confident testimony can be misleading when base rates are ignored.

Probability tree diagram

We can represent the example in a tree diagram.

```
                          Start: Taxi Accident
                                 |



          Blue Taxi (A)                          Black Taxi (~A)
          P(A) = 0.15                           P(~A) = 0.85
              |                                        |


 Says Blue (B)   Says Black (~B)        Says Blue (B)   Says Black (~B)
 P(B|A)=0.80     P(~B|A)=0.20          P(B|~A)=0.20     P(~B|~A)=0.80
       |               |                       |               |

 Joint: 0.12     Joint: 0.03           Joint: 0.17      Joint: 0.68
 P(A B)          P(A ~B)               P(~A B)          P(~A ~B)
```

[1] Another way to see this is, say we present the witness the 30 Blue taxis, she will identify 80% or 24 as blue (and 6 as black) under the same night conditions. Also she will identify 20% or 34 of 170 black taxis as 'blue'. So out of 54 taxis she calls 'blue', only 24 are in fact blue, which is about 41%.

We see that:
$P(AB) = 0.12$
$P(B) = P(A \cap B) + P(\sim A \cap B) = 0.12 + 0.17 = 0.29$ i.e. the marginal probability of B by *integrating out* A by summing across A. And since we are looking for $P(A|B)$ i.e. $P(AB)/P(B) \approx 0.41$, which is the probability the taxi was blue given that the witness says blue!

5.8.0.1 Bayes' Updating

Essentially, we have:

Posterior Probability	$\propto$	Prior Probability	$\times$	Likelihood

Prior Probability (15%): This represents our initial belief about the situation before considering the witness testimony. Based on the town's taxi distribution, we know that only 15% of taxis are blue. This is our baseline expectation that any random taxi involved in an accident would be blue.

Likelihood (Witness Accuracy): This quantifies how probable the observed evidence (witness testimony) is under different hypotheses. The witness correctly identifies blue taxis 80% of the time and incorrectly identifies black taxis as blue 20% of the time. This creates two competing likelihoods: - If the taxi was actually blue, we'd expect her to say "blue" 80% of the time - If the taxi was actually black, we'd expect her to mistakenly say "blue" 20% of the time

Posterior Probability (41%): This is the updated probability that combines our prior belief with the new evidence. Using Bayes' theorem, we find that even though the witness says she saw a blue taxi, there's only about a 41% chance she's actually correct. This counterintuitive result occurs because blue taxis are rare (15%) and the false positives from the much more common black taxis outweigh the true positives.

Intuition Behind the Result: The Bayesian approach forces us to consider both the reliability of the evidence AND the base rates (or prioir provbailities). Even with a seemingly reliable witness (80% accurate), the rarity of blue taxis means most "blue" identifications will be errors. This explains why the court should be skeptical—the witness's claim only increases the probability from 15% to 41%, far from "beyond reasonable doubt."

Extension: What If We Had Another Witness? If a second independent witness also claimed to see a blue taxi, Bayesian updating would apply again. The first posterior (41%) would become our new prior, and we'd combine it with the second witness's reliability. Intuitively, this would increase our confidence, but mathematically, we'd need to consider (1)

whether the witnesses are truly independent, (2) whether they have the same accuracy rates, and (3) the possibility of collusion or shared biases.

With two independent, reliable witnesses both saying "blue," the probability that the taxi was actually blue would increase substantially, potentially making the evidence strong enough for legal purposes. This demonstrates how Bayesian updating elegantly handles accumulating evidence—each new piece of information updates our beliefs in a mathematically rigorous way.

5.9 Axioms of probability

Probability rests on three axioms:

1. **Nonnegativity:** $P(A) \geq 0$
2. **Additivity:** For disjoint events, $P(A \cup B) = P(A) + P(B)$
3. **Normalization:** $P(\mathcal{S}) = 1$

From these axioms follow many useful properties, such as:

- If $A \subset B$, then $P(A) \leq P(B)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cup B) \leq P(A) + P(B)$

! Chapter summary

Probability provides a formal language for reasoning under uncertainty. In this chapter, we introduced experiments, events, conditional probability, independence, and the fundamental rules governing probabilities. These ideas form the foundation for statistical inference in the chapters ahead.

5.10 Exercises

5.10.1 Conceptual questions

1. **Three views of probability**
   Explain the difference between the **classical**, **frequentist**, and **subjective** interpretations of probability.
   Give one real-world example for each interpretation.

2. **Probability without philosophy**
   The chapter argues that the mathematics of probability does not depend on which interpretation we adopt.
   Explain what this means in your own words.

3. **Experiments and outcomes**
   Consider the experiment *"toss a fair coin until a head appears."*

   - What is the sample space?
   - Is this a finite or infinite sample space?
   - Why is this still considered a *single* experiment?

4. **Events vs outcomes**
   Explain the difference between an **outcome** and an **event**.
   Can a single outcome itself be an event?

5. **Conditional probability and information**
   Why does conditional probability capture the idea of *learning from new information*?
   Give an example (not involving cards or coins) where learning new information changes probabilities.

6. **Independence vs mutual exclusivity**
   Are mutually exclusive events ever independent?
   Carefully explain your answer.

5.10.2 Basic probability calculations

7. **Coins and sample spaces**
   A fair coin is tossed four times.

- How many outcomes are in the sample space?
- What is the probability of getting exactly two heads?
- What is the probability of getting at least one head?

8. **Complements**
   Let event A be *"a randomly selected student passes an exam"*, with $P(A) = 0.85$.
   - What is $P(\sim A)$?
   - Interpret this probability in words.
9. **Cards and conditional probability**
   A card is drawn at random from a standard deck.
   - What is the probability that it is a queen?
   - What is the probability that it is a queen **given** that it is a face card?
   - Are the events *"queen"* and *"face card"* independent?

5.10.3 Multiplication rule and independence

10. **Without replacement**
    Two cards are drawn *without replacement* from a standard deck.
    - What is the probability that both cards are red?
    - What is the probability that the first card is an ace and the second card is also an ace?
11. **With replacement**
    Repeat Question 10, but now assume the first card is replaced before the second draw. Explain why the probabilities change.
12. **Testing independence**
    Two events satisfy

    $P(A) = 0.4$, $P(B) = 0.5$, and $P(AB) = 0.2$.
    - Are A and B independent?

- Justify your answer mathematically.

5.10.4 Addition rule and mutually exclusive events

13. **Mutually exclusive or not?**
    For each pair of events below, state whether they are mutually exclusive and explain why:
    - Drawing a heart and drawing a spade
    - Drawing a heart and drawing a red card
    - Getting a head on the first toss and a tail on the second toss
14. **Dice example**
    Two fair dice are rolled.
    - What is the probability that at least one die shows a 6?
    - Use the general addition rule to compute your answer.

5.10.5 Bayes' rule and interpretation

15. **Medical testing**
    A disease affects 2% of a population.
    A diagnostic test correctly identifies the disease 95% of the time, but incorrectly signals disease in 5% of healthy individuals.
    - Define the relevant events.
    - Use Bayes' rule to compute the probability that a person who tests positive actually has the disease.
    - Explain the result in words.
16. **Base-rate neglect**
    In the taxi example discussed in the chapter, many people expect the probability to be close to 0.8.
    Explain why this intuition is misleading.

5.10.6 Set Operations and Probability Identities

17. Set operations and De Morgan's laws

(a) Consider rolling a fair die. Let

- A be the set of outcomes where the roll is an **even number**, and
- B be the set of outcomes where the roll is **greater than 3**.

Calculate and compare the sets on both sides of De Morgan's laws:

$$\sim (A \cup B) = \sim A \cap \sim B \qquad \text{and} \qquad \sim (A \cap B) = \sim A \cup \sim B.$$

(b) Show that

$$\sim A = (\sim A \cap B) \cup (\sim A \cap \sim B),$$

and

$$\sim B = (A \cap \sim B) \cup (\sim A \cap \sim B).$$

(c) Show that

$$\sim (A \cap B) = (\sim A \cap B) \cup (\sim A \cap \sim B) \cup (A \cap \sim B).$$

18. Probability inequality

Prove that for any two events A and B,

$$P(A \cap B) \geq P(A) + P(B) - 1.$$

Hint: Use the axioms of probability and the addition rule.

19. Law of total probability

Show the identity

$$P(A \mid B) = P(C \mid B)P(A \mid B \cap C) + P(\sim C \mid B)P(A \mid B \cap \sim C).$$

Explain briefly how this formula relates to conditioning on whether event C occurs or not.

5.10.7 Reflection and interpretation

20. **Probability and statistical reasoning**
    Explain why probability theory is essential for understanding statistical concepts such as correlation and regression.

21. **From probability to inference**
    Which ideas from this chapter do you think will be most important when we later study statistical inference?
    Briefly explain why.

6 Discrete Probability Distributions

We begin by defining a **random variable** as a function or rule that assigns a number to each outcome of an experiment. Instead of talking about the coin flipping event as {heads, tails}, for example, we can think of it as 1 and 0, what we often call an indicator variable. This is actually a numerical event which can be called 'the number of heads when flipping a coin'.

Formally, a random variable is defined as:

$$X : \Omega \to \mathbb{R}$$

where Ω is the sample space (all possible outcomes of the experiment), X is a measurable function that maps each outcome $\omega \in \Omega$ to a real number $X(\omega)$.

So, the experiment produces an outcome ω, and the random variable translates that outcome into a number. Hence one can view the random variables as a 'bridge' " ' from experiment $\to$ numbers.

The **probability distribution** is a listing of all possible outcomes of an experiment and the corresponding probability. Once we have a random variable X, we can ask: how likely is each possible value of X? That's where the distribution comes in. It is the probability measure induced by X:

$$P_X(A) = P(\{\omega \in \Omega : X(\omega) \in A\})$$

where P is the probability measure on the sample space, P_X is the distribution of X and A is a set of real numbers (e.g., an interval).

This tells us: the probability that X takes values in A is equal to the probability of all outcomes that map into A. In pther words, a distribution is like a rulebook-it specifies the probabilities attached to those numbers, telling us the 'laws' governing randomness.

6.0.1 Tossing a fair coin

This approach shows that correlation:

- Sample space: $\Omega = \{\text{Heads, Tails}\}$
- Random variable: $X(\text{Heads}) = 1$, $X(\text{Tails}) = 0$

- Distribution: $P(X = 1) = 1/2$, $P(X = 0) = 1/2$

A discrete distribution is based on random variables which can assume only clearly separated values i.e. takes on only whole numbers like $0, 1, 2, ...$ while a continuous distribution can assume an infinite number of values i.e. takes on any value on the Real line, within a given range. A useful analogy is "Integers are discrete, while Real Numbers are continuous".

❗ Discrete vs. Continious

- **Discrete random variable:** Distribution given by a probability mass function (PMF)

$$P(X = x_i) = p_i, \quad \sum_i p_i = 1$$

- **Continuous random variable:** Distribution given by a probability density function (PDF)

$$P(a \leq X \leq b) = \int_a^b fx(dx), \quad \int_{-\infty}^{\infty} f(x)dx = 1$$

6.1 The discrete random variable

When dealing with univariate data, we have seen how the histogram together with its mean and standard deviation are a useful way to summarize data. In statistics, what in fact we are dealing with is a **random variable**, sometimes denoted $r.v$, which is nothing more than a function or rule that assigns a number to each outcome of an experiment. And as we have already mentioned, the outcomes of a random variable can be described by a probability distribution, which are a listing of all possible outcomes of an experiment and their corresponding probabilities.

Let us look at an example. This time, consider an experiment in which a coin is **tossed three times**. Let X be a $r.v$ representing the **number of heads in 3 tosses**. The possible outcomes are:

Note we have 8 equally likely outcomes from which we can easily construct the probability distribution of X, the number of heads in 3 tosses, as:

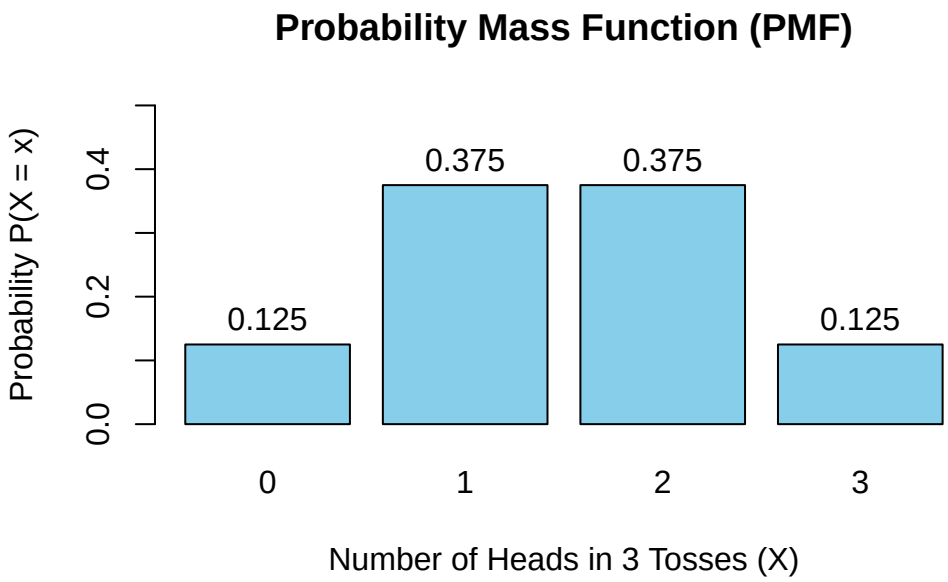
The probability distribution, sometimes also referred to as the probability function, is in fact a histogram which looks as follows:

Table 6.1: Outcomes of 3 tosses of a coin

Outcome	No. of Heads
T T T	0
T T H	1
T H T	1
T H H	2
H T T	1
H H T	2
H T H	2
H H H	3

Table 6.2: Probability distribution of 3 tosses of a coin

X	$P(X)$
0	1/8 or 0.125
1	3/8 or 0.375
2	3/8 or 0.375
3	1/8 or 0.125
	1.000



The corresponding **cumulative probability distribution** can be drawn as:

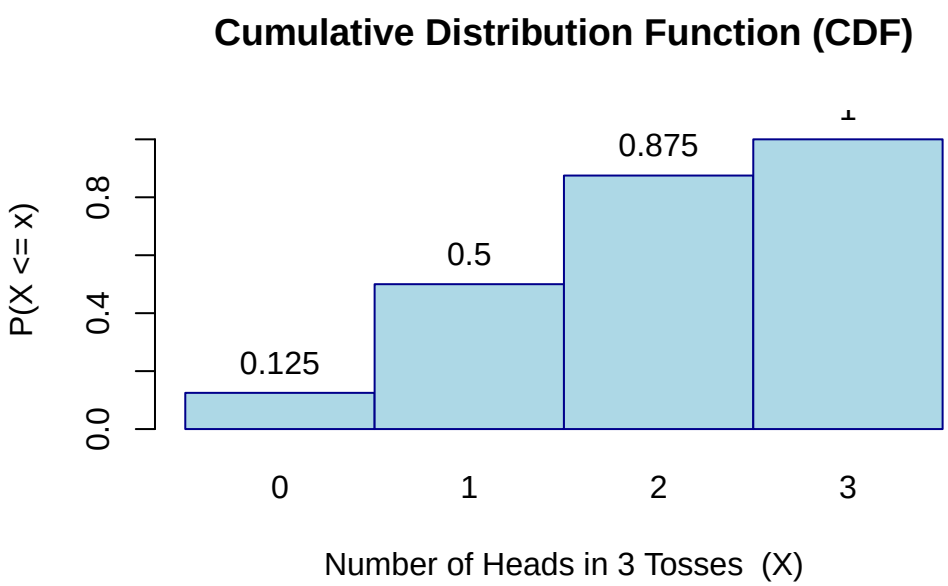


Table 6.3: Calculating the mean.

X	$P(X)$	$X \cdot P(X)$
0	1/8	0
1	3/8	3/8
2	3/8	6/8
3	1/8	3/8
Total	**1**	**12/8 = 1.5**

6.2 The mean and the variance

Apart from drawing the probability distribution or histograms, we have seen that numerical summaries such as the mean and variance are useful.

The mean is the long-run average value of a random variable, which is also referred to as its expected value, denoted $E(X)$.

$$E(X) = \mu = \sum_{i=1}^{n} X_i P(X_i) \tag{6.1}$$

The variance which measures the spread or variation of the distribution is defined as:

$$Var(X) = \sigma^2 = \sum_{i=1}^{n} (X_i - \overline{X})^2 P(X_i) \tag{6.2}$$

To get the standard deviation, simply calculate the square root of the variance.

Let's see this with the example above which defines the random variable X as the number of heads in 3 tosses.

To compute the mean, use Equation 6.1, shown in the table below–summing the last column of the table gives $\mu = 1.5$.

To get the variance, applying Equation 6.2, as shown in the the table below and summing the last column of the table gives $\sigma^2 = 0.75$

Careful, we have the variance $= 0.75$, hence the standard deviation $\sigma = \sqrt{0.75}$ which is about 0.8667.

Table 6.4: Calculating the variance.

X	$P(X)$	$(X_i - \mu)$	$(X_i - \mu)^2 \cdot P(X)$
0	0.125	-1.5	0.28125
1	0.375	-0.5	0.09375
2	0.375	0.5	0.09375
3	0.125	1.5	0.28125
	1.000		$\sigma^2 = \mathbf{0.75}$

6.3 Rules of expected value and variance

Let us introduce some basic rules of expected value and variance. Firstly, for **expected value**, we have:

Rule E1

$$E(k) = k$$

Rule E2

$$E(X + k) = E(X) + k$$

Rule E3

$$E(kX) = kE(X)$$

Rule **E1** states that the expected value (or average) of a constant is that constant. Rule **E2** states that the expected value of a random variable to which a constant has been added is equal to the expected value of the random variable plus the constant. Rule **E3** states that the expected value of a random variable multiplied by a constant is equal to the constant times the expected value of the random variable.

Secondly, the **rules of variance** are:

Rule V1

$$Var(k) = 0$$

Rule V2

$$Var(X + k) = Var(X)$$

Rule V3

$$Var(kX) = k^2 Var(X)$$

Rule **V1** states that the variance of a constant is zero (i.e., there is no spread). Rule **V2** states that the variance of a random variable to which a constant has been added is simply equal to the variance of the random variable. Lastly, Rule **V3** states that the variance of a random

variable multiplied by a constant is equal to the constant squared times the variance of the random variable.^[It is instructive to compare the expected value and variance rules with the summation rules.[1]

Example: Monthly Profit Analysis

Let's illustrate how these rules apply to a real-world scenario.
The Data: For a food store, monthly sales have a mean (μ) of **Baht 25,000** and a standard deviation (σ) of **Baht 4,000**. Profits are defined as 30% of sales minus fixed costs of **Baht 6,000**.
The Goal: Find the mean and standard deviation of the monthly profit.
1. Define the Equation:

$$Profit = 0.30(Sales) - 6,000$$

2. Calculate the Expected Value (Mean):

$$\begin{aligned} V(\text{Profit}) &= V[0.30(\text{Sales}) - 6,000] \\ &= V[0.30(\text{Sales})] && \text{(Rules V2 \& V1)} \\ &= (0.30)^2 V(\text{Sales}) && \text{(Rule V3)} \\ &= 0.09 \cdot (4,000)^2 \\ &= 0.09 \cdot 16,000,000 \\ &= 1,440,000 \end{aligned}$$

3. Calculate the Variance and Standard Deviation: First, we find the variance (V). Remember that $V(Sales) = \sigma^2 = 4,000^2$.

$$\begin{aligned} V(\text{Profit}) &= V[0.30(\text{Sales}) - 6,000] \\ &= V[0.30(\text{Sales})] && \text{(Rules V2 \& V1)} \\ &= (0.30)^2 V(\text{Sales}) && \text{(Rule V3)} \\ &= 0.09 \cdot (4,000)^2 \\ &= 0.09 \cdot 16,000,000 \\ &= 1,440,000 \end{aligned}$$

Finally, the standard deviation is:

$$\sigma_{\text{Profit}} = \sqrt{1,440,000} = \mathbf{1,200}$$

[1] It is instructive to compare the expected value and variance rules with the **summation** rules.

6.4 Discrete bivariate distributions

So far, we have focused on **univariate distributions**. We now extend these ideas to **bivariate probability distributions**, which arise from **joint probabilities**.

A **joint probability distribution** of two discrete random variables X and Y is a table or formula that lists the joint probabilities for all pairs of values (x, y). It is denoted by $P(X, Y)$ or, equivalently, $P(X = x \text{ and } Y = y)$.

As we would expect, joint probabilities satisfy the following basic properties:

$0 \leq P(X, Y) \leq 1$

and

$\sum_x \sum_y P(X, Y) = 1.$

6.4.1 Joint and marginal probabilities

We have already encountered an example of a bivariate probability distribution in the previous chapter. That example involved a survey of university graduates working as CEOs and whether they were **GA graduates**.

The corresponding **joint** and **marginal** distributions are summarized in the table below.

	Becomes CEO	Does not becomes CEO	**Total**
GA graduate	$P(A_1B_1) = 0.11$	$P(A_1B_2) = 0.29$	$P(A_1) = 0.40$
Not GA graduate	$P(A_2B_1) = 0.06$	$P(A_2B_2) = 0.54$	$P(A_2) = 0.60$
Total	$P(B_1) = 0.17$	$P(B_2) = 0.83$	1.00

A_1 is the event that the individual is a GA graduate, or A_2 otherwise.
B_1 is the event that the individual becomes a CEO, or B_2 otherwise.

The **marginal probabilities** are obtained by summing the joint probabilities across rows or down columns. They describe the probabilities of A and B **individually**, ignoring the other variable.

Using marginal probabilities, we can compute the **mean**, **variance**, and **standard deviation** of each variable in a bivariate distribution.

Table 6.6: Marginal Distribution of Event A

A	$P(A)$
1	0.40
0	0.60

Table 6.7: Marginal Distribution of B

B	$P(B)$
1	0.17
0	0.83

6.4.1.1 Marginal distribution of A

Using the expectation Equation 6.1 and variance Equation 6.2 formulas introduced earlier:

- $E(A) = 0.40$
- $Var(A) = 0.24$
- $SD(A) \approx 0.49$

6.4.1.2 Marginal distribution of B

Similarly, let B be an indicator variable that equals 1 if an individual becomes a CEO and 0 otherwise. The marginal distribution of B is:

The corresponding **moments** are:[2]

- $E(B) = 0.17$
- $Var(B) \approx 0.14$
- $SD(B) \approx 0.38$

[2]We will introduce the concept of moments later. Essentially the mean and variance are the first moments and second central moments, respectively.

Order	Raw Moment	Central Moment	Common Name
1st	$\mu_1' = E[X]$	$\mu_1 = E[X - E[X]] = 0$	Mean
2nd	$\mu_2' = E[X^2]$	$\mu_2 = E[(X-\mu)^2]$	Variance
3rd	μ_3'	$\mu_3 = E[(X-\mu)^3]$	Skewness
4th	μ_4'	$\mu_4 = E[(X-\mu)^4]$	Kurtosis

6.4.2 Covariance of two discrete variables

We have already seen that the **covariance** between two discrete random variables A and B is defined as

$$Cov(A, B) = \sum_A \sum_B (A - \overline{A})(B - \overline{B})P(A, B) \tag{6.3}$$

Continuing with the GA–CEO example, the covariance by Equation 6.3 between A and B is approximately

$Cov(A, B) = 0.042$.

6.4.3 Correlation

The **correlation coefficient** is obtained by dividing the covariance by the product of the standard deviations:

$$\text{Corr}(A, B) = \frac{\text{Covariance}(A, B)}{SD_A \, SD_B}$$

For this example, the correlation coefficient $r \approx 0.23$.

We therefore conclude that there is a *weak* **positive relationship** between a GA graduate and becoming a CEO.

6.5 The Binomial formula

We have seen the multiplication and addition rules of probability. Roughly speaking, these rules help us solve "and" and "or" problems. Sometimes, however, we might wish to solve "exact" problems, for example:

1. A coin is tossed 4 times. What is the chance of getting exactly one head?
2. A die is rolled 10 times. What is the chance of getting exactly 3 Aces?
3. A box contains one red marble and nine green ones. Five draws are made at random with replacement. What is the chance that exactly two draws will be red?

It is with such problems that the binomial formula can be very helpful.

Figure 6.1: Box Model

6.5.1 Box Model Example

Let's look more closely at the last example above. We might start with a box model:

From the above, we draw 5 tickets at random with replacement. But we want exactly 2 R's and 3 G's (e.g., $\boxed{\text{R}}, \boxed{\text{R}}, \boxed{\text{G}}, \boxed{\text{G}}, \boxed{\text{G}}$ or $\boxed{\text{G}}, \boxed{\text{R}}, \boxed{\text{G}}, \boxed{\text{G}}, \boxed{\text{R}}$, etc.). The exact probability can be found using 3 simple steps:

1) Find all the possible ways:

$$\frac{5!}{2! \times 3!} = \frac{\text{total cases}}{\text{no. of successes} \times \text{no. of failures}}$$

2) Calculate the chance of each:

$$\frac{1}{10} \times \frac{1}{10} \times \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10} = \left(\frac{1}{10}\right)^2 \left(\frac{9}{10}\right)^3$$

3) Use the addition rule to add up the chances:

$$\frac{5!}{2! \times 3!} \times \left(\frac{1}{10}\right)^2 \left(\frac{9}{10}\right)^3 \approx 7\%$$

This can be interpreted as the chance of getting exactly 2 $\boxed{\text{R}}$'s.

Note that **step 1** is actually the *binomial coefficient*, which gives us the total number of combinations of arranging 2 R's and 3 G's. In **step 2**, the chance of getting $\boxed{\text{R}}, \boxed{\text{R}}, \boxed{\text{G}}, \boxed{\text{G}}, \boxed{\text{G}}$ is the same as getting $\boxed{\text{G}}, \boxed{\text{R}}, \boxed{\text{G}}, \boxed{\text{G}}, \boxed{\text{R}}$, and so on. Lastly, **step 3** involves the addition rules as all events are mutually exclusive (disjoint).

6.5.2 The Binomial Formula

The Binomial Formula

The chance that an event will occur exactly k times out of n trials is given by

$$P(X = k) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}, \text{ for } k = 0, 1, \ldots, n \tag{6.4}$$

Where: - n is the number of trials. - k is the number of times the event of interest is to occur. - p is the probability that the event of interest will occur on any particular trial.

Three conditions must hold for the binomial formula to be valid:

1. The value of n must be fixed in advance.

2. p must be the same from trial to trial.

3. The trials must be independent of each other.

Quick Example

To find the chance of getting exactly 2 aces in 10 rolls of a die:

$$P(X = 2) = \frac{10!}{2!8!} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^8 \approx 0.2907$$

6.6 The Poisson distribution

The Poisson random variable is a discrete random variable that typically fits cases of rare events that occur over a fixed amount of time or within a specified space or region. Typical cases are, for example:

- The number of customers entering a service station per hour.
- The number of errors a typist makes per page.
- The number of telephone calls received by a switchboard per hour.

6.7 The Poisson Formula

The Poisson probability distribution is given by:

$$P(X = x) = f(x) = \frac{e^{-\mu}\mu^x}{x!} \tag{6.5}$$

Where: - μ is the average number of successes in a particular interval of time or space. - e is the constant $\approx 2.71828\ldots$ - x is the number of successes.^[Take care of notation here. We have introduced $f(x)$: the small "f" represents a probability density function.

Correspondingly, a large "F", say, $F(x)$, would refer to the cumulative probability function. This should be distinguished from the large "P" we have been using, which stands for probability.]

6.7.1 Assumptions of the Poisson Process

For the Poisson formula to be valid, the following conditions must be met:

1. **Independence**: The number of successes that occur in one time interval should be independent of the number of successes in another interval.
2. **Proportionality**: The probability of a success in a certain interval should be the same for all intervals of the same size and proportional to the length of the interval.
3. **Simultaneity**: The probability that two or more successes will occur in an interval approaches zero as the interval becomes smaller and smaller.

Example: Typographical Errors

Assume that the number of typographical errors in a new edition of a textbook follows a Poisson distribution with a mean of **1.5 per 100 pages**. If 100 pages are randomly selected, what is the probability that there are no typos ($X = 0$)?

$$\begin{aligned} P(X = 0) = f(0) &= \frac{e^{-1.5}1.5^0}{0!} \\ &= \frac{0.2231 \times 1}{1} \\ &= \mathbf{0.2231} \end{aligned}$$

6.8 Exercises

6.8.1 Understanding discrete random variables

1. **What is a discrete random variable?**
   Explain the difference between a *random outcome* and a *random variable*.
   Give two examples of discrete random variables drawn from everyday life.

2. **Probability distributions**
   A discrete random variable X takes values $x_1, x_2, \dots, x_k$ with probabilities $P(X = x_i)$.

   (a) What two conditions must a probability distribution satisfy?

   (b) Explain in words why these conditions are necessary.

6.8.2 Mean, variance, and transformations

3. **A simple distribution**
   The random variable X has the following distribution:

X	0	1	2	3
$P(X)$	0.4	0.3	0.2	0.1

   (a) Compute $E[X]$, $\mathrm{Var}(X)$, and the standard deviation of X.

   (b) Interpret the mean of X in words.

4. **Linear transformations**
   Suppose $Y = 3X + 2$.

   (a) Use the laws of expectation and variance to compute $E[Y]$ and $\mathrm{Var}(Y)$.

   (b) Find the standard deviation of Y.

   (c) Explain why adding a constant affects the mean but not the variance.

6.8.3 Joint distributions, independence, and correlation

5. **Joint distribution**
   The joint probability distribution of X and Y is given below:

	$Y = -1$	$Y = 0$	$Y = 1$
$X = 0$	0.1	0.1	0.1
$X = 2$	0.1	0.2	0.1

	$Y=-1$	$Y=0$	$Y=1$
$X=4$	0.1	0.1	0.1

(a) Compute the marginal distributions of X and Y.

(b) Show that $E[XY] = E[X]E[Y]$.

(c) Conclude that X and Y are uncorrelated.

(d) Show that X and Y are **not independent**.

Why does zero correlation not imply independence?

6. **Another joint distribution**
The joint distribution of X and Y is given by:

	$X=0$	$X=1$	$X=2$
$Y=1$	1/6	1/6	1/6
$Y=-1$	0	1/2	0

(a) Compute $E[X]$ and $E[Y]$.

(b) Are X and Y independent? Justify your answer.

(c) Compute $\mathrm{Cov}(X, Y)$ and determine whether the variables are correlated.

6.8.4 Counting and discrete probability models

7. **Families and children**
Assume that the probability of a boy equals the probability of a girl, and that births are independent.

(a) In families with **four children**, what proportion have more girls than boys?

(b) List all relevant outcomes explicitly.

8. **Comparing family sizes**
Couple A has **three children**.
Couple B has **five children**.

(a) For each couple, compute the probability that there are more girls than boys.

(b) Is the probability higher for couple A, couple B, or the same?

(c) Explain the intuition behind your result.

6.8.5 Discrete distributions in practice

9. **Guessing on a multiple-choice exam**
A student randomly guesses on a 100-question multiple-choice test. Each question has four possible answers.

   (a) What is the expected number of correct answers?

   (b) What is the standard deviation?

   (c) If a student scores 35 out of 100, would you attribute this more to luck or skill? Explain briefly.

   (d) Which event has higher probability: scoring **exactly 55%**, or scoring **more than 55%**?

10. **Arrivals at an urgent care center**
The mean number of arrivals at an urgent care facility is 4 per hour.

(a) Assuming arrivals follow a Poisson process, what is the probability of exactly 2 arrivals in one hour?

(b) Explain why the Poisson distribution is appropriate in this context.

11. **Website traffic**
Hits to a personal website occur randomly and independently, with an average of 4 hits per week.

(a) What is the probability of receiving 10 or more hits in one week?

(b) What is the probability of receiving 20 or more hits over two weeks?

12. **Manufacturing defects**
Flaws in a carpet occur randomly at a rate of one per 300 square feet.

(a) What distribution is appropriate for modeling the number of flaws?

(b) What is the probability that an 8×10 foot carpet contains **no flaws**?

6.8.6 Conceptual wrap-up

13. **Big picture**

(a) How do discrete probability distributions help us model uncertainty?

(b) Why are expectations and variances central to economic analysis?

(c) Give one example from economics where a discrete random variable is more appropriate than a continuous one.

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